Statistical Analysis of Physical Characteristics of Palmer Penguins

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Objective

Briefly state that the aim is to determine whether **body mass and flipper length differ across penguin species and sex** using statistical analysis.

Dataset Description

- Palmer Penguins dataset
- 344 penguin observations
- Species: Adelie, Chinstrap, Gentoo
- Variables used:
  - species
  - sex
  - body_mass_g
  - flipper_length_mm

Dataset Cleaning

Before performing statistical analysis, missing observations were removed for the variables required for each individual analysis.

For body mass analysis:

```
body_mass_data <- penguins %>%
  filter(!is.na(body_mass_g))
```

For the male-versus-female analysis:

```
sex_data <- penguins %>%
  filter(!is.na(body_mass_g), !is.na(sex))
```

For flipper-length analysis:

```
flipper_data <- penguins %>%
filter(!is.na(flipper_length_mm))
```

Missing values were not replaced with artificial values because doing so could influence the statistical results.

Descriptive Statistics

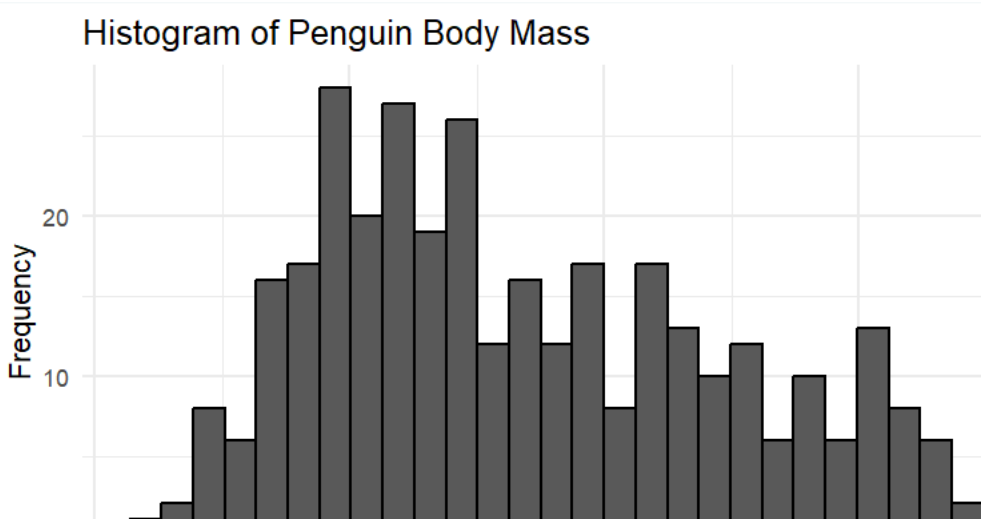
1. Overall Body Mass

The descriptive statistics of body mass were calculated after excluding the two missing body-mass observations.

Statistic	Body Mass
Number of observations	**342**
Mean	**4201.75 g**
Median	**4050 g**
Minimum	**2700 g**
Maximum	**6300 g**
Q1	**3550 g**
Q3	**4750 g**
IQR	**1200 g**

The overall body mass ranges from 2700 g to 6300 g, with an average of approximately 4202 g.

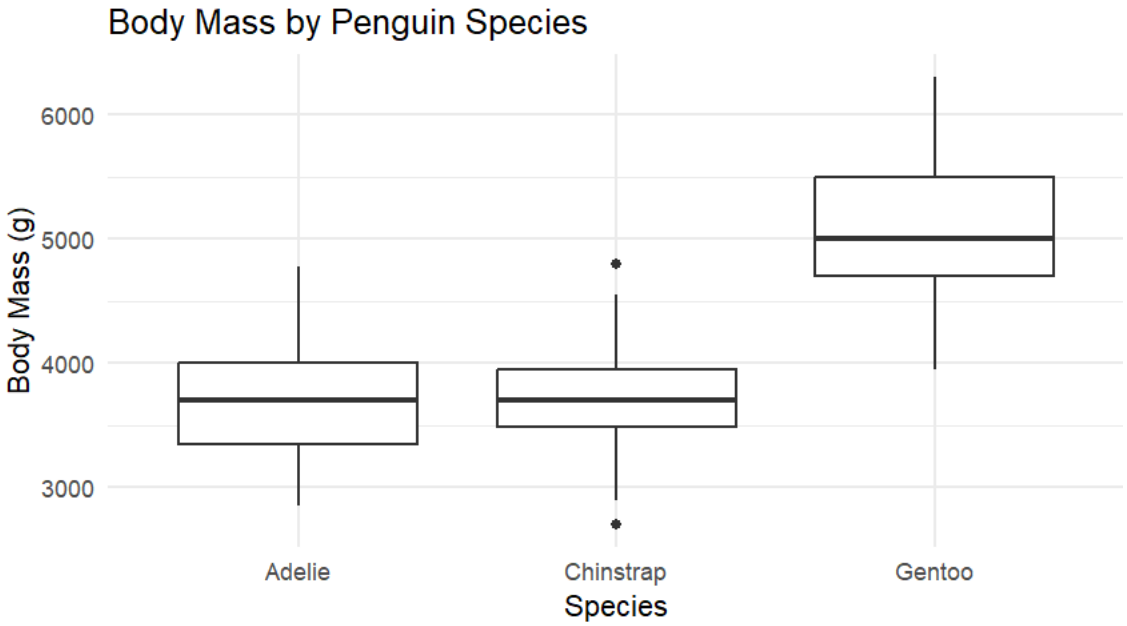
The difference between the mean and median indicates that the distribution is not perfectly symmetric.



2. Body Mass by Species

Species	n	Mean (g)	Median (g)	SD (g)	Min (g)	Max (g)
Adelie	151	3700.66	3700	458.57	2850	4775
Chinstrap	68	3733.09	3700	384.34	2700	4800
Gentoo	123	5076.02	5000	504.12	3950	6300

Gentoo penguins have substantially higher body mass than Adelie and Chinstrap penguins. Adelie and Chinstrap penguins have very similar average body masses.



3. Skewness and Kurtosis

The distribution of body mass was also examined using skewness and kurtosis.

Species	Skewness	Kurtosis
Adelie	0.2825	2.4056
Chinstrap	0.2419	3.4637
Gentoo	0.0688	2.2579

The positive skewness values indicate slight right-skewness. Gentoo has the smallest skewness and therefore has a comparatively more symmetric distribution.

4. Descriptive Visualization

The following graphs were used:

1. Histogram of body mass.
2. Boxplot of body mass by species.
3. Density plot of body mass.
4. Species-wise density plot.
5. Boxplot of body mass by sex.

The visualizations show a clear separation of Gentoo penguins from Adelie and Chinstrap penguins in terms of body mass.

Hypothesis testing: Male vs Female Body Mass

1. Objective

An independent two-sample t-test was performed to determine whether the mean body mass differs between male and female penguins.

Null Hypothesis

H_o: There is no significant difference between the mean body mass of male and female penguins.

[
H_0:\mu_{male}=\mu_{female}
]

Alternative Hypothesis

H_1: There is a significant difference between the mean body mass of male and female penguins.

[

H_1:\mu_{male}\neq\mu_{female}

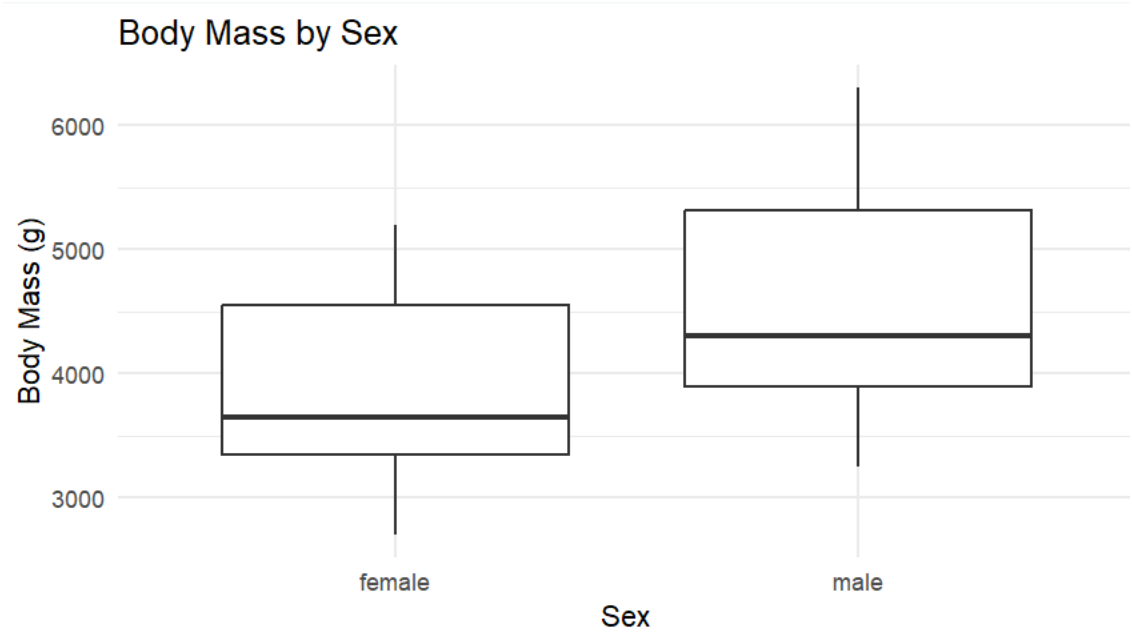
]

2. Mean Body Mass by Sex

Sex	Mean Body Mass
Female	**3862.27 g**
Male	**4545.68 g**

Male penguins have a considerably higher average body mass than female penguins. The difference between the means is approximately:

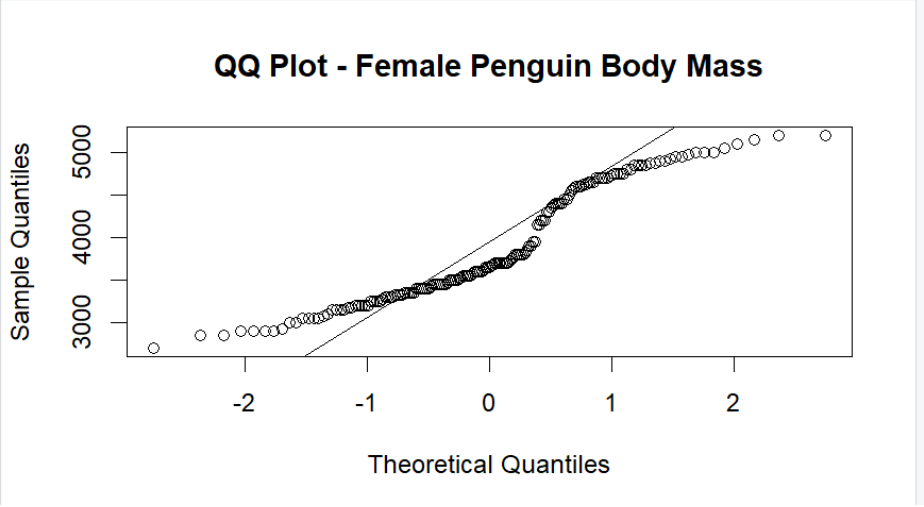
4545.68 – 3862.27 = 683.41 g



3. Normality Test

Shapiro-Wilk tests and Q-Q plots were used to examine the normality assumption.

The Q-Q plots were approximately linear, supporting the use of the t-test for comparing the two groups.



4. Independent Two-Sample t-Test

An independent two-sample Welch t-test was performed because the two groups have different sample sizes and potentially different variances.

The test evaluates whether the observed difference of approximately 683 g could have occurred by chance.

The resulting p-value is **far below 0.05**, so the null hypothesis is rejected.

Interpretation

There is statistically significant evidence that male and female penguins differ in body mass.

Male penguins are significantly heavier than female penguins.

5. 95% Confidence Interval

The 95% confidence interval for the difference in mean body mass does not include zero.

Therefore, the confidence interval supports the conclusion obtained from the t-test.

6. Effect Size

Cohen's d was calculated to determine the magnitude of the difference.

The effect size indicates a **large practical difference** between male and female body mass.

Therefore, the difference is not only statistically significant but also practically meaningful.

One-Way ANOVA

1. Objective

   A one-way ANOVA was performed to determine whether mean body mass differs among Adelie, Chinstrap and Gentoo penguins.

Null Hypothesis

[
H_0:\mu_{Adelie}=\mu_{Chinstrap}=\mu_{Gentoo}
]

Alternative Hypothesis

At least one species has a different mean body mass.

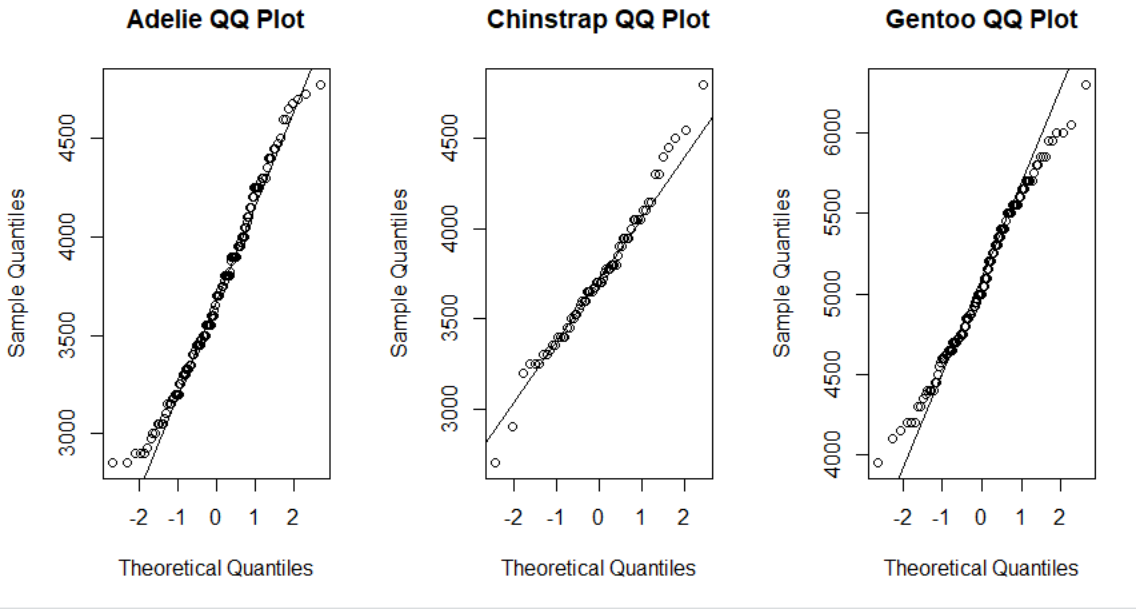
2. Normality

Shapiro-Wilk tests were performed separately for each species.

Species	Shapiro-Wilk p-value	Conclusion
Adelie	0.0324	Significant deviation
Chinstrap	0.5605	Approximately normal
Gentoo	0.2336	Approximately normal

The Adelie group shows a statistically significant departure from perfect normality, while Chinstrap and Gentoo do not.

Because the sample sizes are reasonably large, ANOVA is relatively robust to moderate departures from normality.



3. Levene's Test

Levene's test was used to test equality of variances.

F(2,339) = 5.3355

p = 0.00523

Since p < 0.05, the equal-variance assumption is violated.

Therefore, the results should be interpreted together with the non-parametric Kruskal-Wallis analysis.

4. One-Way ANOVA Test

The ANOVA produced:

F(2,339) = 343.6

$p < 2.2 \times 10^{-16}$

The result is statistically significant.

Conclusion

The null hypothesis is rejected.

There is strong evidence that body mass differs among the three penguin species.